Motion, forces and energy

90 questions: 45 multiple choice, 45 structured.

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1 · Part 1

Core | Physical quantities and measurement techniques | [1]

A student places a pencil along a ruler. One end of the pencil is level with the 2.0 cm mark. The other end is level with the 15.5 cm mark.

What is the length of the pencil?

A. 15.5 cm

B. 13.5 cm

C. 17.5 cm

D. 14.5 cm

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2 · Part 1

Core | Physical quantities and measurement techniques | [1]

A student uses a stopwatch to time 20 complete oscillations of a pendulum. The stopwatch reads 31.6 s.

What is the period of the pendulum?

A. 0.63 s

B. 15.8 s

C. 31.6 s

D. 1.58 s

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3 · Part 1 Core Physical quantities and measurement techniques [1]

A measuring cylinder contains water. The water level reads 50.0 cm^3. A student lowers 40 identical small ball bearings into the water. They are all completely covered. The water level now reads 54.8 cm^3.

What is the volume of one ball bearing?

A 0.12 cm^3

B 1.4 cm^3

C 4.8 cm^3

D 8.3 cm^3

4 · Part 1 Extended Physical quantities and measurement techniques [1]

Which list contains only vector quantities?

A force, mass, velocity

B speed, acceleration, weight

C momentum, weight, gravitational field strength

D distance, velocity, energy

5 · Part 1 Extended Physical quantities and measurement techniques [1]

Car X travels due north at 15 m / s. Car Y travels due east at 15 m / s.

Which statement is correct?

A The cars have the same speed but different velocities.

B The cars have the same velocity but different speeds.

C The cars have the same speed and the same velocity.

D The cars have different speeds and different velocities.

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6 · Part 1 Extended Physical quantities and measurement techniques [1]

Two horizontal cables pull on a floating buoy. One cable pulls with a force of 7.0 N due west. The other cable pulls with a force of 24 N due south.

What is the size of the resultant of these two forces?

A 17 N

B 25 N

C 31 N

D 625 N

7 · Part 1 Core Motion [1]

A horse and rider travel 4.5 km in 15 min at a constant speed.

What is their speed in m / s?

A 0.30 m / s

B 300 m / s

C 18 m / s

D 5.0 m / s

8 · Part 1 Core Motion [1]

A bus travels 12 km along a road at a constant speed of 40 km / h. It then travels a further 12 km at a constant speed of 60 km / h.

What is the average speed of the bus for the whole 24 km?

A 50 km / h

B 24 km / h

C 48 km / h

D 100 km / h

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9 · Part 1 Core Motion [1]

The table shows the distance travelled by a toy robot moving in a straight line.

time / s	0	2.0	4.0	6.0	8.0
distance / m	0	6.0	12.0	12.0	12.0

The robot moves steadily between each pair of readings. Which statement describes the motion of the robot?

A It accelerates and then is at rest.

B It moves at a constant speed and then is at rest.

C It moves at a constant speed for the whole 8.0 s.

D It accelerates and then moves at a constant speed.

10 · Part 1 Core Motion [1]

A stone is dropped from rest near the surface of the Earth. Air resistance on the stone is negligible.

Which statement about the acceleration of the stone as it falls is correct?

A It is approximately constant, at about 9.8 m / s^2.

B It increases steadily as the stone falls faster.

C It is approximately constant, at about 9.8 m / s.

D It decreases to zero as the stone falls.

11 · Part 1 Extended Motion [1]

A motorbike travelling in a straight line slows down steadily from 24 m / s to 9.0 m / s in 30 s.

What is the acceleration of the motorbike?

A +0.50 m / s^2

B −0.30 m / s^2

C −1.1 m / s^2

D −0.50 m / s^2

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12 · Part 1 Extended Motion [1]

The table gives the speed, in m / s, of four objects at intervals of 1.0 s. Each object moves in a straight line.

Which object is moving with a changing acceleration?

	t = 0	t = 1.0 s	t = 2.0 s	t = 3.0 s
A	2.0	4.0	6.0	8.0
B	5.0	5.0	5.0	5.0
C	0	1.0	4.0	9.0
D	12.0	9.0	6.0	3.0

13 · Part 1 Extended Motion [1]

A raindrop falls vertically through still air at its terminal velocity.

Which statement about the raindrop is correct?

A The air resistance is equal in size to the weight, and the acceleration is zero.

B The air resistance is zero, and the acceleration is 9.8 m / s^2.

C The air resistance is greater than the weight, and the raindrop is slowing down.

D The air resistance is less than the weight, and the speed is increasing.

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14 · Part 1 Core Motion [1]

The distance–time graph for a remote-controlled car is a straight line between t = 2.0 s and t = 6.0 s. The table gives two points on this straight-line section.

time / s	**2.0**	**6.0**
distance / m	7.0	19.0

What is the speed of the car during this section?

A 3.5 m / s

B 3.0 m / s

C 3.2 m / s

D 0.33 m / s

15 · Part 1 Core Motion [1]

An electric scooter starts from rest. Its speed increases steadily to 6.0 m / s in 8.0 s. It then travels at a constant 6.0 m / s for a further 5.0 s.

How far does the scooter travel in the whole 13.0 s?

A 78 m

B 48 m

C 54 m

D 39 m

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16 · Part 1 Core Physical quantities and measurement techniques [4]

A student wants to find the thickness of one card from a pack of 50 identical playing cards. She stacks all 50 cards neatly and stands a ruler upright beside the stack. The bottom of the stack is level with the 1.0 cm mark. The top of the stack is level with the 2.6 cm mark.

(a) Suggest why the student measures the whole stack instead of measuring one card. [1]

(b) Determine the height of the stack. [1]

(c) Calculate the thickness of one card. Give your answer in mm. [2]

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17 · Part 1 Core Physical quantities and measurement techniques [6]

A student investigates a pendulum made from a small mass on a string. She uses a stopwatch to time 10 complete oscillations. She does this three times. Her results are 16.4 s, 16.8 s and 16.6 s.

(a) Explain why the student times 10 oscillations instead of one oscillation. [1]

(b) Calculate the period of the pendulum. [2]

(c) Suggest one way the student could make her starting and stopping of the stopwatch more accurate. [1]

(d) The student also wants to measure the time taken for a toy car to pass through a 5.0 cm gap. This time is about 0.1 s.

Suggest why a hand-operated stopwatch is not suitable for this measurement, and describe a better method. [2]

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18 · Part 1 Extended Physical quantities and measurement techniques [4]

(a) State the difference between a scalar quantity and a vector quantity. [1]

(b) Here is a list of quantities.

distance, momentum, temperature, electric field strength, speed, acceleration

Identify which of these quantities are vectors and which are scalars. [2]

(c) A student says, "Mass and weight are both measured when you stand on bathroom scales, so they must both be scalars."

Explain why the student is wrong about weight. [1]

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19 · Part 1 Extended Physical quantities and measurement techniques [4]

Two people pull a heavy box across a smooth floor using horizontal ropes. One rope pulls with a force of 72.0 N due east. The other rope pulls with a force of 135 N due north. The two ropes are at right angles to each other.

(a) Calculate the magnitude of the resultant of these two forces. [2]

(b) Calculate the angle between the resultant force and the 72.0 N force. [2]

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20 · Part 1 Extended Physical quantities and measurement techniques [6]

A delivery drone flies with a velocity of 12 m / s due north relative to the air. A steady wind blows due east. The resultant velocity of the drone relative to the ground has a magnitude of 13 m / s.

(a) Calculate the speed of the wind. [2]

(b) Calculate the angle between the direction of the resultant velocity and due north. [2]

(c) A student says that the speed of the drone relative to the ground should be 12 m / s + the wind speed.

Explain why the student is wrong. [2]

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21 · Part 1 Core Motion [6]

A delivery van makes a journey in three stages.

- Stage 1: it travels 18 km in 20 min.
- Stage 2: it stops at a depot for 10 min.
- Stage 3: it travels 27 km in 30 min.

(a) Define speed. [1]

(b) State how velocity is different from speed. [1]

(c) Calculate the average speed of the van for the whole journey. Give your answer in km / h. [3]

(d) The van's speed during stage 1 and during stage 3 is 54 km / h.

Explain why the average speed for the whole journey is less than 54 km / h. [1]

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22 · Part 1 Core Motion [7]

A jogger runs along a straight path. The table shows the distance she has travelled from her starting point at different times.

time / s	0	10	20	30	40	50	60	70
distance / m	0	30	60	90	90	90	135	180

Her distance–time graph is made of straight-line sections joining these points.

(a) State what the jogger is doing between t = 30 s and t = 50 s. Explain how the table shows this. [2]

(b) Calculate her speed between t = 0 and t = 30 s. [2]

(c) Calculate her speed between t = 50 s and t = 70 s. [1]

(d) Calculate her average speed for the whole 70 s. [2]

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23 · Part 1 Core Motion [7]

A lift in a tall building travels upwards from rest. Its speed–time graph is made of three straight lines:

- from (0 s, 0 m / s) to (4.0 s, 2.5 m / s)
- from (4.0 s, 2.5 m / s) to (16.0 s, 2.5 m / s)
- from (16.0 s, 2.5 m / s) to (21.0 s, 0 m / s)

(a) Describe the motion of the lift during each of the three stages. [2]

(b) Calculate the distance the lift travels in the first 4.0 s. [2]

(c) Calculate the total distance the lift travels in the 21.0 s. [3]

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24 · Part 1 Core Motion [5]

A small steel ball is released from rest and falls vertically near the surface of the Earth. Air resistance is negligible. A data logger records the speed of the ball.

time / s	**0**	**0.10**	**0.20**	**0.30**	**0.40**
speed / m / s	0	0.98	1.96	2.94	3.92

(a) State whether the ball is at rest, moving at a constant speed or accelerating. Give a reason from the data. [2]

(b) State the approximate value of the acceleration of free fall near the surface of the Earth. [1]

(c) The speed–time graph for the ball is a straight line through the origin.

Calculate the distance the ball falls in the first 0.40 s. [2]

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25 · Part 1 Extended Motion [6]

A skateboarder moves along a straight, flat path. The table shows her speed at different times.

time / s	0	3.0	6.0	9.0	12.0	15.0
speed / m / s	2.0	5.0	8.0	8.0	4.0	0

Her speed–time graph is made of straight-line sections joining these points.

(a) Define acceleration. [1]

(b) Calculate her acceleration between t = 0 and t = 6.0 s. [2]

(c) Calculate her acceleration between t = 9.0 s and t = 15.0 s. [2]

(d) State her deceleration between t = 9.0 s and t = 15.0 s. [1]

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26 · Part 1 Extended Motion [6]

A sports car starts from rest and moves along a straight test track. The table shows its speed at intervals of 1.0 s.

time / s	0	1.0	2.0	3.0	4.0	5.0
speed / m / s	0	6.0	11.0	15.0	18.0	20.0

(a) Determine whether the car moves with a constant acceleration or a changing acceleration. Explain your answer using the data. [2]

(b) Calculate the average acceleration of the car over the first 5.0 s. [2]

(c) Deduce the 1.0 s interval in which the average acceleration of the car is greatest, and calculate this acceleration. [2]

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27 · Part 1 Extended Motion [5]

An astronaut standing on the Moon holds a hammer and a feather at the same height. She releases them at the same moment. The Moon has no atmosphere.

(a) State and explain what the astronaut observes. [2]

(b) A student repeats the experiment on the Earth, in air.

Describe how the result is different from the result on the Moon, and explain why. [2]

(c) State what is meant by terminal velocity. [1]

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28 · Part 1 Extended Motion [6]

A train leaves a station from rest along a straight track. It accelerates at a constant 0.60 m / s^2 for 40 s. Later, it approaches the next station at this same top speed and decelerates at a constant 0.80 m / s^2 until it stops.

(a) Calculate the top speed of the train. [2]

(b) Calculate the time the train takes to stop. [2]

(c) Calculate the distance the train travels while it is decelerating. [2]

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29 · Part 1 Core Motion [6]

Two cyclists, P and Q, ride in the same direction along a long, straight cycle path. The table shows the distance of each cyclist from a signpost at different times. Each cyclist moves at a constant speed.

time / s	**0**	**5.0**	**10.0**	**15.0**	**20.0**
distance of P / m	0	30	60	90	120
distance of Q / m	50	70	90	110	130

(a) Calculate the speed of cyclist P. [2]

(b) Calculate the speed of cyclist Q. [1]

(c) Both cyclists keep moving at the same constant speeds after t = 20.0 s.

Determine the time at which cyclist P catches up with cyclist Q. [2]

(d) A third cyclist R rides at the same speed as P, but in the opposite direction along the path.

State whether R has the same velocity as P. Give a reason. [1]

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30 · Part 1 Extended Motion [5]

A small metal ball is released from rest at the surface of a tall tube of thick syrup. It falls vertically through the syrup. The table shows its speed at different times.

time / s	**0**	**0.10**	**0.20**	**0.30**	**0.40**	**0.50**	**0.60**
speed / m / s	0	0.42	0.68	0.83	0.90	0.90	0.90

(a) Determine whether the acceleration of the ball during the first 0.30 s is constant or changing. Use the data to justify your answer. [2]

(b) State the terminal velocity of the ball. [1]

(c) Explain, in terms of the forces on the ball, why it reaches a terminal velocity. [2]

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1 · Part 2 Core Mass and weight [1]

An engineer packs a toolbox on Earth. The toolbox is later used on Mars, where the gravitational field strength is smaller than on Earth.

Which statement about the toolbox on Mars is correct?

A Its mass and its weight are both smaller than on Earth.

B Its mass is the same as on Earth and its weight is smaller than on Earth.

C Its mass is smaller than on Earth and its weight is the same as on Earth.

D Its mass and its weight are both the same as on Earth.

2 · Part 2 Core Mass and weight [1]

A rock sample has a mass of 2.5 kg. On the surface of a distant dwarf planet, the weight of the sample is 1.5 N.

What is the gravitational field strength on the surface of the dwarf planet?

A 1.7 N / kg

B 3.8 N / kg

C 9.8 N / kg

D 0.60 N / kg

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3 · Part 2 Core Mass and weight [1]

Two bags of rice, P and Q, are placed on opposite pans of an equal-arm beam balance. The beam comes to rest level.

Which statement about P and Q while they are on the balance is correct?

A P and Q have equal weights, so they also have equal masses.

B P and Q have equal masses, but their weights could be different.

C P and Q have equal weights, but their masses could be different.

D The balance shows only that P and Q have equal volumes.

4 · Part 2 Core Density [1]

A small block of an alloy has a mass of 540 g and a volume of 60 cm^3.

What is the density of the alloy?

A 0.11 g / cm^3

B 9.0 kg / m^3

C 9.0 g / cm^3

D 32 400 g / cm^3

5 · Part 2 Core Density [1]

The density of water is 1.0 g / cm^3. The table gives the mass and the volume of four solid objects. None of the objects absorbs water.

Which object floats in water?

	mass / g	**volume / cm^3**
A	120	160
B	40	25
C	90	75
D	260	200

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6 · Part 2 Extended Density [1]

Three liquids, X, Y and Z, do not mix with each other.

- The density of X is 0.92 g / cm^3.
- The density of Z is 1.00 g / cm^3.
- A 50 cm^3 sample of Y has a mass of 63 g.

The liquids are poured gently into a tall glass container in the order Z, then Y, then X. They are left to settle into three layers.

Which list gives the order of the layers from the top of the container to the bottom?

A Y, Z, X

B X, Z, Y

C Z, X, Y

D X, Y, Z

7 · Part 2 Core Effects of forces [1]

A sledge is pulled across flat snow. Two ropes pull the sledge towards the east with forces of 120 N and 45 N. A friction force of 30 N acts on the sledge towards the west. All three forces act along the same straight line.

What is the resultant force on the sledge?

A 195 N towards the east

B 165 N towards the east

C 135 N towards the west

D 135 N towards the east

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8 · Part 2 Core Effects of forces [1]

A cyclist rides around a circular track at a constant speed.

Which statement is correct?

A There is no resultant force on the cyclist, because the speed is constant.

B There is a resultant force on the cyclist, so the speed must be changing.

C There is a resultant force on the cyclist, because the direction of motion is changing.

D There is no resultant force on the cyclist, because the cyclist is moving.

9 · Part 2 Core Effects of forces [1]

Which statement about friction is correct?

A Drag acts on an object moving through air, but not on an object moving through water.

B Drag acts on an object moving through water and on an object moving through air.

C Friction between two solid surfaces always helps the surfaces to slide past each other.

D Friction between two solid surfaces sliding past each other makes the surfaces colder.

10 · Part 2 Extended Effects of forces [1]

An unloaded spring has a length of 12.0 cm. When a load of 6.0 N hangs from the spring, its length is 16.0 cm. The spring has not passed its limit of proportionality.

What is the spring constant of the spring?

A 1.5 N / cm

B 0.38 N / cm

C 0.67 N / cm

D 24 N / cm

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11 · Part 2 Extended Effects of forces [1]

A trolley of mass 2.5 kg moves along a horizontal bench. A forward force of 8.0 N acts on the trolley, and a friction force of 3.0 N acts in the opposite direction.

What is the acceleration of the trolley?

A) 3.2 m / s^2

B) 4.4 m / s^2

C) 12.5 m / s^2

D) 2.0 m / s^2

12 · Part 2 Extended Effects of forces [1]

A car travels around a circular bend on a flat road. The force that keeps the car moving in a circle is the friction force between the tyres and the road. This force acts perpendicular to the car's direction of motion.

Which change would need a larger friction force to keep the car moving in a circle?

A) The car carries fewer passengers, at the same speed, around a circle of the same radius.

B) The car travels more slowly, with the same passengers, around a circle of the same radius.

C) The car carries more passengers, at the same speed, around a circle of the same radius.

D) The car travels at the same speed, with the same passengers, around a circle of larger radius.

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13 · Part 2 Core Turning effect of forces [1]

A mechanic tightens a nut with a spanner. She applies a force of 40 N to the spanner, at right angles to the spanner, at a perpendicular distance of 0.25 m from the centre of the nut.

What is the moment of this force about the centre of the nut?

A) 10 Nm

B) 160 Nm

C) 0.0063 Nm

D) 1000 Nm

14 · Part 2 Core Turning effect of forces [1]

A seesaw is pivoted at its centre. A child of weight 300 N sits 1.6 m to the left of the pivot. A second child, of weight 400 N, sits on the right of the pivot so that the seesaw balances. Ignore the weight of the seesaw.

How far from the pivot does the second child sit?

A 2.1 m

B 1.2 m

C 1.6 m

D 0.75 m

15 · Part 2 Extended Turning effect of forces [1]

A light horizontal beam is pivoted at its centre. All the forces on the beam act vertically downwards.

- On the left of the pivot: a 20 N force acts 0.40 m from the pivot, and a 15 N force acts 0.20 m from the pivot.
- On the right of the pivot: a 30 N force acts 0.30 m from the pivot.

What downward force must act on the right of the pivot, 0.50 m from the pivot, so that the beam balances?

A 22 N

B 18 N

C 4.0 N

D 5.0 N

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16 · Part 2 Core Mass and weight [5]

A crate of emergency supplies has a mass of 25 kg. The gravitational field strength on Earth is 9.8 N / kg.

(a) State what is meant by the mass of an object. [1]

(b) State what is meant by the weight of an object. [1]

(c) Define gravitational field strength. [1]

(d) Calculate the weight of the crate on Earth. [2]

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17 · Part 2 Core Mass and weight [7]

A space agency is testing a robotic lander of mass 1200 kg. The table gives the gravitational field strength at the surface of three bodies in the Solar System.

body	gravitational field strength / (N / kg)
Earth	9.8
Mars	3.7
Titan	1.4

(a) Calculate the weight of the lander on the surface of Mars. [2]

(b) On the surface of one of the other bodies in the table, the weight of the lander is 1680 N. Calculate the gravitational field strength at the surface of this body and identify the body. [3]

(c) State the acceleration of free fall near the surface of Mars. [1]

(d) State the mass of the lander when it is on the surface of Titan. [1]

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18 · Part 2 Extended Mass and weight [6]

A scientist has a mineral sample of mass 0.75 kg. On Earth, where the gravitational field strength is 9.8 N / kg, she measures the sample in two ways:

- by hanging it from a newtonmeter (a spring balance that reads force in N)
- by placing it on one pan of an equal-arm beam balance and adding standard masses to the other pan until the beam is level.

She repeats both measurements on the Moon, where the gravitational field strength is 1.6 N / kg.

(a) Describe, in terms of the gravitational field, why the sample has weight. [1]

(b) Calculate the reading on the newtonmeter on Earth and on the Moon. [2]

(c) On the Moon, the beam balance is level with the same standard masses as on Earth, but the newtonmeter reading is different. Explain these two observations. [3]

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19 · Part 2 Core Density [7]

A student finds the density of a small irregularly shaped stone. The stone sinks in water.

- The mass of the stone, measured on a balance, is 64.8 g.
- A measuring cylinder contains water. The reading on the measuring cylinder is 42.0 cm^3.
- The stone is lowered into the water until it is completely covered. The reading on the measuring cylinder rises to 66.0 cm^3.

(a) Calculate the volume of the stone. [1]

(b) Calculate the density of the stone. [2]

(c) Describe how the student could use a balance and an empty measuring cylinder to determine the density of a liquid such as cooking oil. [3]

(d) State one precaution the student should take when reading the measuring cylinder. [1]

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20 · Part 2 Core Density [7]

A solid plastic block is a rectangular block measuring 5.0 cm × 4.0 cm × 2.0 cm. Its mass is 32 g. The plastic does not absorb any liquid.

(a) Calculate the volume of the block. [1]

(b) Calculate the density of the plastic. [2]

(c) The density of water is 1.0 g / cm^3. The density of a liquid L is 0.70 g / cm^3. Determine whether the block floats or sinks in water and whether it floats or sinks in liquid L. Give a reason for each answer. [2]

(d) A small piece of metal is fixed firmly to the outside of the block. The piece of metal has a mass of 15 g and a volume of 2.0 cm^3. Determine whether the block and metal together float or sink in water. [2]

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21 · Part 2 Core Effects of forces [5]

A delivery van moves forwards in a straight line along a level road. At one moment, three horizontal forces act on the van:

- a forward driving force of 2400 N
- air resistance of 900 N, acting backwards
- a rolling friction force of 300 N, acting backwards.

(a) Calculate the resultant force on the van and state its direction. [2]

(b) State what happens to the speed of the van at this moment. [1]

(c) Later, the van moves at a constant speed in a straight line. The driving force is now 2000 N and the rolling friction force is still 300 N. Determine the air resistance on the van. Explain your reasoning. [2]

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22 · Part 2 Core Effects of forces [6]

A kayaker paddles a kayak across a lake. At the end of the trip, she drags the kayak up a sandy beach.

(a) State the type of friction that acts on the kayak as it moves through the water. [1]

(b) When the kayak is dragged across the sand, the underside of the kayak becomes warm. Describe what is meant by solid friction and use your description to explain why the underside becomes warm. [2]

(c) On the lake, the kayaker stops paddling. Explain why the kayak slows down. [2]

(d) Suggest one way in which the kayaker could reduce the friction on the kayak when she moves it across the beach. [1]

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23 · Part 2 Core Effects of forces [5]

A tennis ball travels horizontally towards a player. The player hits the ball with a racket. The ball leaves the racket travelling in the opposite direction, faster than it arrived.

(a) While the racket is in contact with the ball, the ball is squashed. State the effect of a force that this shows. [1]

(b) A resultant force can change the velocity of an object in two different ways. State these two ways, and state which of them happen to the tennis ball while it is in contact with the racket. [3]

(c) An astronaut in deep space, far from any planet or star, throws a ball. No forces act on the ball after it leaves her hand. Describe the motion of the ball after it leaves her hand. [1]

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24 · Part 2 Extended Effects of forces [8]

A student hangs loads from a spring and measures the length of the spring for each load. The table shows her results.

load / N	length / cm
0	8.0
1.0	10.5
2.0	13.0
3.0	15.5
4.0	18.0
5.0	21.1
6.0	25.0

(a) Define the spring constant. [1]

(b) Use the results for loads from 0 to 4.0 N to calculate the spring constant of the spring. [2]

(c) State what is meant by the limit of proportionality. Use the results to deduce the two loads in the table between which the limit of proportionality is reached. [3]

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(d) Calculate the length of the spring when the load is 2.6 N. [2]

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25 · Part 2 Extended Effects of forces [6]

A rider on an electric scooter moves along a straight, level path. The total mass of the rider and scooter is 80 kg. At one moment, the forward force from the motor is 340 N and the total resistive force is 100 N.

(a) Calculate the resultant force on the rider and scooter. [1]

(b) Calculate the acceleration of the rider and scooter. [2]

(c) State the direction of this acceleration. [1]

(d) Later, while the scooter is moving forwards, the motor is switched off and the rider brakes. The total backward force on the rider and scooter is then 560 N. Calculate the size of the acceleration and state its direction. [2]

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26 · Part 2 Core Effects of forces [6]

A model rocket is launched vertically upwards. At one instant during the launch, the mass of the rocket is 0.50 kg and it is moving upwards. At this instant:

- the upward force from the rocket motor is 12 N
- the air resistance on the rocket is 2.0 N.

The gravitational field strength is 9.8 N / kg.

(a) Calculate the weight of the rocket at this instant. [2]

(b) Determine the size and the direction of the resultant force on the rocket at this instant. [2]

(c) Later, the motor stops but the rocket is still moving upwards. State the direction of the resultant force on the rocket now, and state its effect on the speed of the rocket. [2]

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27 · Part 2 Core Turning effect of forces [6]

A person opens a door by pushing on its handle. The door turns about its hinges.

(a) Describe what is meant by the moment of a force, and give one everyday example of a moment other than opening a door. [2]

(b) Define the moment of a force about a pivot. [1]

(c) The person pushes the handle with a force of 15 N, at right angles to the door. The handle is 0.80 m from the hinges. Calculate the moment of this force about the hinges. [2]

(d) Suggest why door handles are placed as far as possible from the hinges. [1]

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28 · Part 2 Core Turning effect of forces [6]

A plank rests horizontally on a pivot at its centre. A box of weight 150 N is placed on the plank 1.4 m to the left of the pivot. A worker pushes vertically downwards on the plank at a point 1.5 m to the right of the pivot, so that the plank stays balanced and horizontal. Ignore the weight of the plank.

(a) Calculate the moment of the weight of the box about the pivot. [2]

(b) Calculate the force with which the worker pushes down on the plank. [2]

(c) The plank is in equilibrium. State the two conditions that must be true for an object to be in equilibrium. [2]

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29 · Part 2 Extended Turning effect of forces [7]

A gardener holds a loaded wheelbarrow still, with its legs just off the ground. The axle of the wheel acts as the pivot. All the forces are vertical, and all the distances below are horizontal distances from the axle.

- The weight of the empty wheelbarrow is 150 N and acts 0.40 m from the axle.
- The weight of the sand in the wheelbarrow is 500 N and acts 0.36 m from the axle.
- The gardener lifts the handles with a vertical upward force F, 1.20 m from the axle.
- The ground pushes vertically upwards on the wheel, directly below the axle.

The weights of the wheelbarrow and the sand act on the same side of the axle as the handles.

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(a) Calculate the total moment of the two weights about the axle. [2]

(b) Calculate the force F. [2]

(c) Calculate the upward force of the ground on the wheel. [2]

(d) Suggest how the gardener could load the same sand so that a smaller force F is needed. [1]

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30 · Part 2 Extended Turning effect of forces [6]

A student is given a metre rule, a pivot on which the rule can balance, and a set of weights that can be hung from the rule.

(a) Describe an experiment the student could do with this apparatus to demonstrate that there is no resultant moment on an object in equilibrium. [4]

(b) In one test, the rule balances horizontally on the pivot at its centre. A 2.0 N weight hangs 0.300 m to the left of the pivot and a 1.5 N weight hangs 0.400 m to the right of the pivot. Use calculations to justify the statement that there is no resultant moment on the rule. [2]

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1 · Part 3 Core Centre of gravity [1]

Which statement describes the centre of gravity of an object?

A the point at the geometric centre of the object, whatever its shape

B the lowest point of the object when it rests on a surface

C the point inside the object where its density is greatest

D the point through which all of the weight of the object appears to act

2 · Part 3

Core

Centre of gravity

[1]

The desk lamp in a student's room has a flat circular base and a tall vertical stem. The lamp falls over too easily when it is knocked.

Which change makes the lamp more stable?

A make the base wider and fix a heavy metal disc inside the base

B make the base narrower but keep the mass of the lamp the same

C fix a heavy weight near the top of the stem

D make the stem taller but keep the base the same

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3 · Part 3

Core

Centre of gravity

[6]

In a science lesson, a student has a flat piece of stiff card with an irregular shape. She wants to find the position of its centre of gravity.

She has a clamp and stand holding a horizontal pin, a plumb line (a thread with a small heavy mass tied to the end), a pencil and a sharp point for making holes in the card.

(a) State what is meant by the centre of gravity of an object. [1]

(b) Describe how the student uses this apparatus to find the position of the centre of gravity of the card. [4]

(c) The student makes a third hole in the card, hangs the card from it and draws a third line in the same way. State how this third line shows that her result is correct. [1]

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4 · Part 3 Extended Momentum [1]

The total mass of a cyclist and her bicycle is 85 kg. They travel along a straight road at a velocity of 6.0 m / s.

What is the momentum of the cyclist and bicycle?

A 14 kg m / s

B 91 kg m / s

C 510 kg m / s

D 1530 kg m / s

5 · Part 3 Extended Momentum [1]

On a straight, level track, trolley X of mass 2.0 kg moves at 3.0 m / s towards trolley Y of mass 1.0 kg. Trolley Y is at rest. The trolleys collide and stick together. Friction is negligible.

What is the speed of the two trolleys just after the collision?

A 1.5 m / s

B 2.0 m / s

C 3.0 m / s

D 6.0 m / s

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6 · Part 3

Extended

Momentum

[6]

In a game, a ball of mass 0.27 kg travels horizontally at 8.0 m / s towards a vertical wall. It hits the wall and bounces straight back along the same line at 6.0 m / s. The ball is in contact with the wall for 0.090 s.

(a) Calculate the change in momentum of the ball. [3]

(b) State the impulse that the wall exerts on the ball. Give the unit. [1]

(c) Calculate the average force that the wall exerts on the ball. [2]

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7 · Part 3 Extended Momentum [6]

Two ice skaters, P and Q, stand still facing each other on smooth ice. They push against each other's hands and move apart in a straight line. Friction from the ice is negligible.

Skater P has a mass of 64 kg. Skater Q has a mass of 48 kg. Just after the push, skater P moves at 1.5 m / s.

(a) Define momentum. [1]

(b) State the total momentum of the two skaters just before they push. [1]

(c) Calculate the speed of skater Q just after the push. [3]

(d) State the direction in which skater Q moves, compared with skater P. [1]

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8 · Part 3 Extended Momentum [7]

In a car safety test, a crash-test dummy of mass 80 kg is moving forwards at 15 m / s. During the crash, an airbag brings the dummy to rest in 0.15 s.

(a) Define impulse. [1]

(b) Calculate the change in momentum of the dummy. [2]

(c) Calculate the average resultant force on the dummy while the airbag stops it. [2]

(d) In a second test there is no airbag. The dummy, moving at the same speed, is stopped by hitting a hard surface in a much shorter time.

Explain, in terms of momentum, why the force on the dummy is larger in the second test. [2]

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9 · Part 3 Core Energy [1]

Which list contains only energy stores?

A heating, kinetic, chemical

B sound, light, nuclear

C electric current, gravitational potential, internal (thermal)

D kinetic, chemical, elastic (strain)

10 · Part 3 Core Energy [1]

When a small battery-powered fan is switched on, its blades start to spin.

Which statement describes the main energy transfer?

A Energy is transferred from the chemical store of the battery to the kinetic store of the blades by an electric current.

B Energy is transferred from the kinetic store of the blades to the chemical store of the battery by heating.

C Energy is transferred from the chemical store of the battery to the gravitational potential store of the blades by heating.

D Energy is transferred from the internal (thermal) store of the battery to the kinetic store of the blades by light.

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11 · Part 3 Extended Energy [1]

In an experiment, a ball of mass 0.50 kg is released from rest and falls vertically through 3.2 m. Air resistance is negligible. The acceleration of free fall g is 9.8 m / s^2.

What is the speed of the ball after it has fallen 3.2 m?

A 5.6 m / s

B 7.9 m / s

C 31 m / s

D 63 m / s

12 · Part 3 Core Energy [4]

In a short time, a television is supplied with 900 J of energy by an electric current. The table shows what happens to this energy.

energy transfer	energy / J
by light from the screen	135
by sound from the speakers	45
by heating the television and its surroundings	?

(a) State the principle of the conservation of energy. [1]

(b) Calculate the energy transferred by heating the television and its surroundings. [2]

(c) Name the energy store that increases because of the energy transfer in (b). [1]

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13 · Part 3 Extended Energy [7]

In a skate park, a skateboarder starts from rest at the top of a curved ramp. The top of the ramp is a vertical height of 2.5 m above the bottom. The total mass of the skateboarder and skateboard is 50 kg. The acceleration of free fall g is 9.8 m / s^2.

(a) Calculate the decrease in gravitational potential energy as the skateboarder moves from the top to the bottom of the ramp. [2]

(b) Assume that no energy is transferred to other stores. Calculate the speed of the skateboarder at the bottom of the ramp. [3]

(c) In fact, the speed of the skateboarder at the bottom of the ramp is 5.0 m / s. Calculate the energy transferred to other stores, such as the internal (thermal) store of the surroundings, as the skateboarder moves down the ramp. [2]

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14 · Part 3 Extended Energy [6]

On a building site, a crane is driven by a diesel engine. The engine turns a winch, which pulls a cable to lift a crate. The table shows the energy for each of the two stages while the crate is lifted.

stage	energy input / kJ	useful energy output / kJ	energy wasted / kJ
engine	2000	600	?
winch and cable	600	?	120

The useful energy output of the engine is the energy input to the winch and cable. The useful energy output of the winch and cable is transferred to the gravitational potential store of the crate.

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(a) Determine the two missing values in the table. [2]

(b) State the energy store into which most of the wasted energy is finally transferred. [1]

(c) Calculate the overall efficiency of the crane, from the diesel fuel to the gravitational potential store of the crate. [2]

(d) Suggest one way of reducing the energy wasted in the winch and cable. [1]

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15 · Part 3 Core Energy [4]

In a classroom, a student stretches an elastic band and uses it to launch a small paper pellet vertically upwards. The pellet rises, stops for an instant at its highest point, then falls back to the floor and comes to rest.

(a) State the energy store that holds the energy just before the band is released. [1]

(b) Describe the energy transfers from the moment the band is released until the pellet reaches its highest point. [2]

(c) State where the energy has ended up once the pellet has landed and stopped moving. [1]

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16 · Part 3 Core Work [1]

In a supermarket, a shopper pushes a trolley with a constant horizontal force of 35 N. The trolley moves 12 m in the direction of the force.

How much work does the shopper do on the trolley?

A 0.34 J

B 2.9 J

C 47 J

D 420 J

17 · Part 3 Core Work [5]

In a hotel, a lift is raised by a steel cable. The total weight of the lift and its passengers is 6400 N. The cable raises the lift vertically upwards through 15 m at a constant speed.

(a) State how the work done on the lift by the cable is related to the energy transferred to the lift. [1]

(b) Calculate the work done on the lift by the cable as it rises 15 m. [2]

(c) The lift then rises further at the same constant speed. The cable does another 48 000 J of work on the lift. Calculate the further distance the lift rises. [2]

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18 · Part 3 Core Energy resources [1]

In a gas-fired power station, natural gas is burned to generate electrical power.

In which order does the energy pass through these three parts of the power station?

A turbine → boiler → generator

B generator → boiler → turbine

C boiler → turbine → generator

D boiler → generator → turbine

19 · Part 3 Core Energy resources [1]

Which energy resource is renewable but is not reliable, because it is not always available when it is needed?

A wind

B geothermal

C natural gas

D nuclear fuel

20 · Part 3 Extended Energy resources [1]

For most energy resources, radiation from the Sun is the main original source of the energy.

For which energy resource is this not true?

A coal

B hydroelectric

C wind

D tidal

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21 · Part 3 Extended Energy resources [1]

The input power of a lamp is 300 W. Every minute, the lamp transfers 4.5 kJ of energy by light. This is its useful energy output.

What is the efficiency of the lamp?

A 1.5%

B 25%

C 75%

D 400%

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22 · Part 3 Core Energy resources [6]

The people of a mountain region get their electrical power from a hydroelectric power station. A river has been blocked by a dam, and water is stored in a lake behind the dam.

(a) Describe how the hydroelectric power station generates electrical power from the water stored behind the dam. [3]

(b) Give one advantage and one disadvantage of the hydroelectric power station compared with a power station that burns natural gas. [2]

(c) Engineers suggest building wind turbines in the region instead. Give one disadvantage of wind turbines compared with the hydroelectric power station. [1]

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23 · Part 3 Core Energy resources [5]

(a) Describe how electrical power is generated from geothermal resources. [3]

(b) Explain why only a small number of countries generate a large amount of electrical power from geothermal resources. [1]

(c) A house has solar panels and solar cells on its roof. State one difference between what solar panels do and what solar cells do. [1]

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24 · Part 3 Extended Energy resources [6]

(a) State the process by which energy is released in the Sun. [1]

(b) Scientists are carrying out research into using the process in (a) to generate electrical power. State whether this process is used at present to generate electrical power on a large scale. [1]

(c) A nuclear power station uses nuclear fuel. Its reactor transfers energy to the boiler at a rate of 3000 MW. The electrical power output of the power station is 990 MW.

(i) Calculate the efficiency of the power station. [2]

(ii) Calculate the power wasted by the power station. [1]

(iii) Suggest where most of the wasted energy goes. [1]

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25 · Part 3 Extended Energy resources [6]

On a garden shed there is a panel of solar cells. The panel has an area of 1.6 m^2. On a sunny day, each square metre of the panel receives 1000 W of power from sunlight. The electrical power output of the panel is 240 W.

(a) Calculate the total power of the sunlight received by the panel. [1]

(b) Calculate the efficiency of the panel. [2]

(c) Calculate the electrical energy output of the panel in 2.0 hours. [2]

(d) Explain why the efficiency of the panel can never be greater than 100%. [1]

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26 · Part 3 Core Power [1]

The weight of a student is 520 N. She runs up a flight of stairs and rises through a vertical height of 3.5 m in 4.0 s.

What is the useful power she develops in raising her weight?

A 460 W

B 1800 W

C 2100 W

D 7300 W

27 · Part 3 Core Pressure [1]

Carrying a heavy rucksack is more comfortable when it has wide shoulder straps instead of narrow ones.

Why is this?

A The force on the shoulders is smaller.

B The pressure on the shoulders is smaller because the area of contact is larger.

C The pressure on the shoulders is larger because the area of contact is larger.

D The weight of the rucksack is smaller because it is spread out.

28 · Part 3 Extended Pressure [1]

In the sea, a diver swims down from a depth of 4.0 m to a depth of 19 m below the surface of the sea. The density of sea water is 1030 kg / m^3. The gravitational field strength g is 9.8 N / kg.

What is the increase in the pressure on the diver?

A 1.5×10^4 Pa

B 4.0×10^4 Pa

C 1.5×10^5 Pa

D 1.9×10^5 Pa

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29 · Part 3 Core Pressure [6]

The dimensions of a rectangular concrete block are 0.50 m × 0.40 m × 0.30 m. Its weight is 2400 N. The block rests on flat ground on its smallest face.

(a) Define pressure. [1]

(b) Calculate the pressure that the block exerts on the ground. [3]

(c) The block is turned over so that it rests on its largest face. State and explain how the pressure on the ground changes. [2]

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30 · Part 3 Extended Pressure [6]

The water butt in a garden is a large tank that collects rainwater. A tap is fitted near the bottom of the tank, 1.8 m below the surface of the water. The density of water is 1000 kg / m^3. The gravitational field strength g is 9.8 N / kg.

(a) Describe how the pressure beneath the surface of a liquid depends on:

(i) the depth below the surface [1]

(ii) the density of the liquid. [1]

(b) Calculate the pressure at the tap caused by the water above it. Ignore the pressure of the atmosphere. [2]

(c) The opening of the tap is closed by a flat cap of area 2.0 cm^2. Calculate the force of the water on the cap. Ignore the pressure of the atmosphere. [2]

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